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LETTER

Optical-wireless cooperative sleep control with traffic prediction for energy efficient radio access network

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Abstract While 6G necessitates more stringent performance requirements, efforts towards carbon neutrality are prompting a focus on energy-saving in networks. However, the power consumption of distributed units (DUs) in radio access network (RAN) is expected to significantly increase due to the rise in computational load for antenna signals processing resulting from rising traffic volumes. Thus, we propose an energy-saving method for RAN that sleeps unused DUs and optical transceivers using traffic prediction with machine learning. Our evaluation showed that the power consumption reduction effect can be increased more than 20% compared to implementing only DU sleep.
Keywords: radio access network, energy saving, machine learning, traffic prediction
Classification: Network

1. Introduction

While the various efforts are being made to achieve carbon neutrality worldwide, it is necessary to reduce power consumption while satisfying strict performance requirements such as extremely high-speed, high-capacity, and low-latency communication in 6G. In the access network, wired and wireless parts are predicted to respectively consume 4% and 96% of the total power consumption by 2030. Therefore, it is important to reduce the power consumption of the radio access network (RAN). In the RAN, base stations account for 80% of the power consumption [1], prompting the consideration of energy-saving methods that involve putting base stations to sleep.

The Third Generation Partnership Project (3GPP) provides an energy-saving method that compares the traffic load information of each cell with a threshold and puts the cell to sleep when the load is below the threshold [2]. In addition, a method for energy-saving control has been proposed in which base stations with low utilization are put to sleep using traffic prediction [3, 4]. These methods utilize parameters such as the signal-to-interference-plus-noise ratio (SINR) to determine which neighboring cell a user equipment (UE) under a sleeping base station should hand over to. If the parameter values meet certain conditions, the UE will change its target cell. An issue common to all of these methods is that when the UE changes its connection, the wireless environment, including the distance to the antenna and the presence of obstacles, changes, which can lead to increased delays and degraded wireless transmission performance. Furthermore, in areas where there are relatively few base stations, such as rural areas, there are few adjacent base stations and no base station to which the UE can change the connection destination, so this cannot be applied. To solve this problem, it is necessary to conserve part of the base station's power while keeping the antenna part operational in order to maintain the wireless transmission performance. In 5G, to achieve flexibility and efficiency, the base station is divided into centralized unit (CU), distributed unit (DU), and radio unit (RU), with the DU responsible for processing radio signals. As traffic volume continues to increase in the future, it will be necessary to increase the throughput of the RAN. To achieve this, generating radio signals that require complex signal processing is necessary, but this will increase the computational load in the DU's signal processing and result in higher power consumption. Therefore, we need to study methods for energy-saving in RAN, especially with regard to the DU. To this end, we assume a network configuration in which the all-photonics network (APN) [5] is applied to a mobile fronthaul (MFH) to further improve the energy-saving effects. The APN can provide extremely large-capacity, ultra-reliable, and low-latency communication and improve the RAN energy-saving effects by eliminating electrical processing from conventional networks.

For energy-saving in the DU, we previously proposed an energy-saving method that aggregates traffic and puts unused DUs to sleep during low load conditions, and applied a method that utilizes traffic prediction to set the maximum traffic volume of DUs as a sleep control threshold to further improve energy efficiency [6]. By putting the DU to sleep while the RU is operating, it is possible to save power without degrading the wireless transmission performance. Furthermore, since the UE does not need to change the base station to which it is connected, the application range of the proposed method is not limited.

In this paper, in order to further improve the energy-saving effects, we extend the application range of sleep control of our proposed method by making both DU and optical transport network sleep targets. High energy-saving effects can be expected by wider sleep control while maintaining the wireless transmission performance. We report the results of simulations and experiments evaluating the energy-saving effects on the DUs and optical transport network. In this study, we assume that many DUs are put to sleep during the

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time when traffic volume is low at night, and perform sleep control in several hours. Therefore to show the effectiveness of the proposed method, we used a typical traffic pattern of high traffic volume during the day and low traffic volume at night. The simulations utilized a RAN simulator to emulate 5G new radio transmission, and the experiments were conducted with layer 2 (L2) switches and transponders.

2. Proposed method

Figure 1 shows the network configuration of the proposed method. We assume that APN is applied to the MFH. The DU and RU are connected to the APN via transponders, while the controller is connected to the DU through extended cooperative transport interface (eCTI), which is defined as a cooperative interface between RAN and APN [7]. The controller has collection, analysis, and control functions for the sleep control decision based on acquired traffic via eCTI. The collection function gathers information about the traffic volume for each DU. The analysis function executes traffic prediction based on the above collected information and determines which DUs to sleep/wake by comparing the threshold for the sleep control decision with the current/predicted total traffic of multiple DUs. Moreover, due to the route switching between RU and DU that occurs with the sleep/wake of DUs, it is necessary to identify which routes to switch. This process also involves identifying transceivers that become unused as a result of the route switching. The control function executes sleep control for the target sleep/wake DU and the unused transponders' transceivers, as well as optical path switching control between the DU and the RU, based on the decision result of the analysis function. In Fig. 1, traffic is aggregated to one DU, and the unused DUs are in sleep mode. Additionally, as a result of putting the DU in sleep mode, route switching occurs, leading to the presence of unused transceivers. These unused transceivers are put under sleep control and are referred to as "sleeping transceivers," and there are a total of six sleeping transceivers in Fig. 1. The transceivers that remain in use after the route switching are referred to as "active transceivers."

Figure 2 (a) shows an overview of DU sleep control within the analysis function. First, the current traffic volume per DU is summed and compared to a threshold. Second, traffic prediction for each DU is performed using a prepared model, with the results compared to the threshold. If both values are below the threshold, the DU sleeps. By confirming both values, we can avoid frequent DU sleep/wake iterations due to sudden traffic changes. Furthermore, we can use the DU's maximum traffic volume as the threshold, and the threshold need not have a margin for sudden traffic fluctuations.

Figure 2 (b) shows the flowchart for the sleep control in the analysis function. First, a prediction model is constructed using traffic volume information over a certain period. The current traffic volume T_n for each DU is collected via eCTI, and the traffic volumes from multiple DUs are summed up and compared to a threshold S. Next, using the constructed prediction model, the traffic prediction for each DU is performed, and the predicted traffic results T_{n_pred} for multiple DUs are summed and compared to S. When adding traffic, it should be accumulated only up to the point where it does not exceed the threshold. Based on these results, the decision is made to put the DU to sleep when both values are less than S. By aggregating traffic to the DU with the highest traffic volume as the traffic aggregation point, the processing related to rerouting and other activities resulting from traffic aggregation is reduced. When the DU to be put into sleep mode is determined, it is decided which route between the RU and DU needs to be changed. As a result

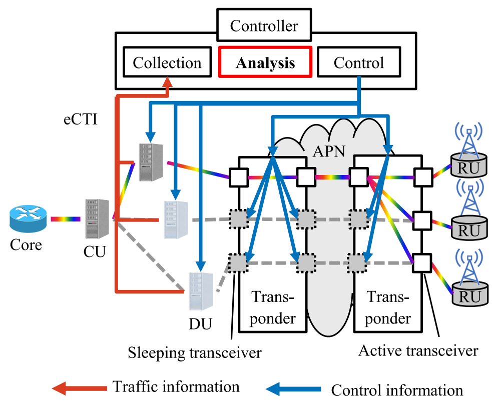

Fig. 1 Network configuration of proposed method.

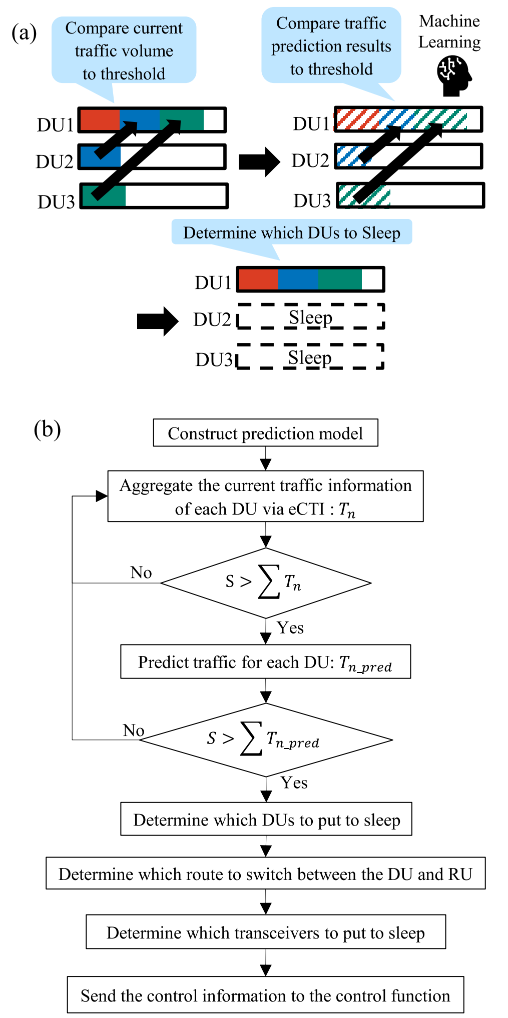

Fig. 2 (a) Overview and (b) flowchart of DU sleep control.

of route determination, any unused transceivers can also be targeted for sleep. Finally, the information for sleep control and route switching control in the optical transport network, as determined by the analysis function, is sent to the control function. In the case of waking from sleep, the current traffic volume and the predicted traffic volume of the DU where traffic is aggregated are compared to the threshold. If both values exceed the threshold, it is decided to wake the DU. Similarly to sleep control, after deciding which DU to wake up, the route switching between the RU and DU and the transceivers to be woken up are determined.

3. Experimental results

3.1 Experimental setup

The experimental setup is shown in Fig. 3. We evaluated the power consumption of the DU in the simulation and that of the optical transport network in the experiment.

In this paper, the power consumption of the DU was re-evaluated using the VIAVI RIC tester, which can more accurately emulate wireless environments compared to previous network simulators. The RAN simulation scenario shown in Fig. 3 (a) involves a macro cell site consisting of three sectors containing three macro cells and an RU installed at the center of these three cells. For each macro cell, three

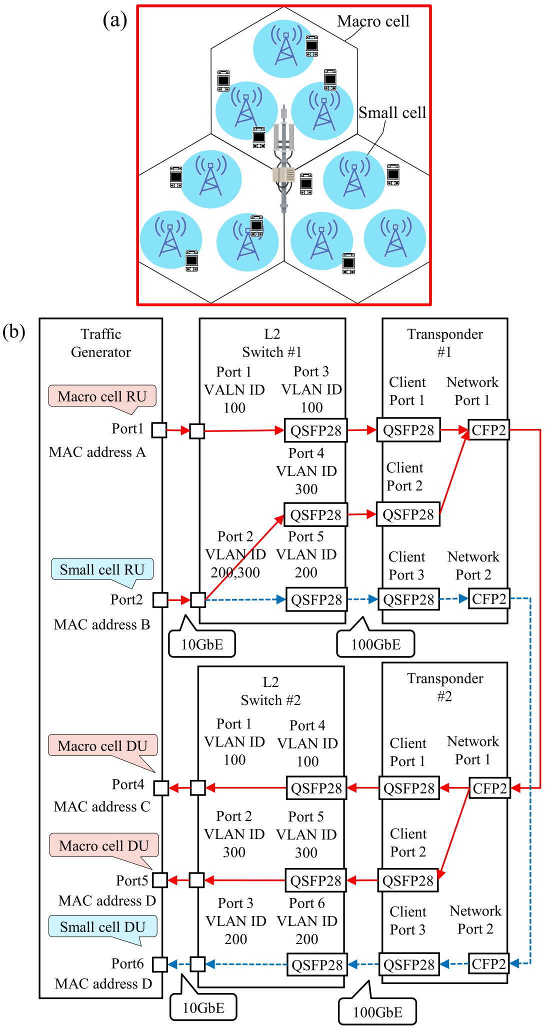

Fig. 3 (a) RAN simulation scenario and (b) experimental setup.

small cells were placed with an RU installed at the center of each small cell [8]. The total number of DUs was ten, because one DU accommodated three macro cells while the number of DUs was one per small cell. In this study, we created traffic patterns based on weekly traffic trends of mobile communications provided by the Ministry of Internal Affairs and Communications [9], assuming a typical pattern of high traffic volume during the day and low volume at night. We used the VIAVI RAN simulator and varied the number of UEs between ten and 30, distributing them randomly. We collected the hourly downlink traffic volume for each cell and created a 24-hour traffic pattern based on this information. The radio center frequency of the macro cell was 4.5 GHz and that of the small cell was 3.7 GHz, and the radio bandwidth was 100 MHz. We selected Transformer as the algorithm to construct the prediction model and used 24 hours of data to predict one hour ahead. The threshold for the sleep control decision was 434.24 Mbps, which is the maximum traffic volume for DU in this simulation. We assume the power consumption of a DU was 300 W [10] and evaluated power consumption for 24 hours. In our proposed method, a predictive model is created in the analysis function of the controller, and traffic prediction is performed every hour. Although this predictive process results in some increase in power consumption, the reduction achieved by putting many DUs to sleep is expected to exceed this, making the increase negligible when considering the whole system.

In the optical transport network experiment, we evaluated the power consumption of the network by the sleep control of one DU. When the small cell DU was put to sleep, we performed route switching, assuming the traffic of a small cell RU was aggregated to a macro cell DU with multiple ports, and evaluated the power consumption reduction using L2 switches (Cisco Catalyst 9500) and transponders (Wistron Galileo Flex T). Wavelength multiplexing devices such as Reconfigurable Optical Add-Drop Multiplexers (ROADMs) were excluded because they do not have transceivers, and a large energy-saving effect is not expected. In the experimental setup shown in Fig. 3 (b), the red and blue lines represent the traffic flow when a DU is and is not put to sleep, respectively. We represented the medium access control (MAC) addresses with alphabets for convenience and defined them as A, B, C, and D for the five ports of the traffic generator to emulate the traffic between the RUs and DUs of macro and small cells. The virtual local area network (VLAN) identifier (ID) for each port of L2 switches #1 and #2 was set as shown in Fig. 3 (b). A VLAN ID of 100 was assigned to the packets of the macro cell RU. Packets of the small cell were assigned VLAN ID 200 before the route switching and VLAN ID 300 after the route switching. With this VLAN ID setting, the VLAN ID of the packets can be identified in the L2 switches and the route can be switched. In the client ports of transponders #1 and #2, 100G quad small form-factor pluggable 28 (QSFP28) transceivers were applied. In the network ports in transponders #1 and #2, we used 400G centrum gigabit form-factor pluggable 2 (CFP2) digital coherent optics (DCO) transceivers and set the wavelength with a frequency of 192.1 THz for network port 1 and 192.2 THz for network port 2.

Before route switching, L2 switch #1 receives the macro cell packets on port 1 and forwards them to port 3, while the small cell packets are received on port 2 and forwarded to port 5. Transponder #1 sends these to network ports, and transponder #2 returns them to client ports. L2 switch #2 forwards the macro cell packets from port 4 to port 1, and the small cell packets from port 6 to port 3. After route switching, L2 switch #1 receives the small cell packets on port 2 and forwards them to port 4. Through transponders #1 and #2, L2 switch #2 receives the small cell packets on port 5 and forwards them to port 2. The macro cell's packets were forwarded to the same port as before the route switch.

Due to the route switching, the QSFP28 of the L2 switch and the QSFP28 and CFP2-DCO of the transponder become unused, enabling a reduction in power consumption by putting these transceivers to sleep.

3.2 Results

Figure 4 shows the evaluation results for the power consumption of the DU. Under these conditions, we found that two DUs of the small cells slept and the maximum of eight DUs were operating from 20:00 to 22:00. This period had the highest traffic volume within the 24-hour period. Since the power consumption of one DU is assumed to be 300 W, the total power consumption was 2400 W when eight DUs were operating. When there was one DU in operation at 4:00, we can achieve a reduction in power consumption of 2100 W compared to having eight DUs operating, since seven DUs of the small cell can sleep. The proposed method thus demonstrated its effectiveness by allowing DUs to be put to sleep not only at night but also during the day.

Table I lists the experimental results for the power consumption of the optical transport network when one small cell DU is put to sleep. By examining the differences in power consumption in Table I, it is evident that a total of 65 W of power can be saved in the optical transport network, with the CFP2-DCO of transponders #1 and #2 accounting for the majority of the power consumption. The CFP2-DCO is capable of high-speed data transmission and long-distance transmission by digital coherent technology, and the power consumption is higher than that of the QSFP28. If only the DU is considered, the power consumption reduction is 300 W. However, when the optical transport network is also considered, the power consumption reduction becomes 300 + 65 = 365 W, resulting in a 21.7% improvement in energy-saving efficiency.

Based on the results from Fig. 4 and Table I, assuming that all the DUs are connected to the same L2 switch and transponder, the power consumption can be reduced by $(300 + 65) \times 7 = 2555$ W when seven DUs sleep. Therefore, we confirmed the improvement in energy-saving effectiveness through the proposed method, resulting in energy-efficient RAN.

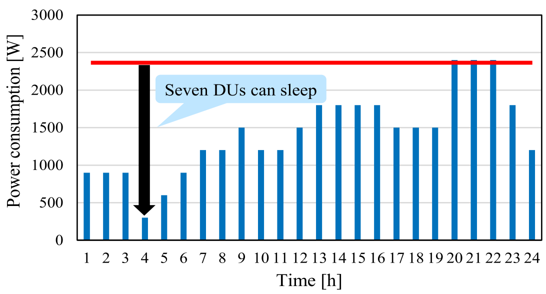

Fig. 4 Power consumption of DU.

Table I Power consumption of the optical transport network when one DU is put to sleep.

	Not sleep [W]	Sleep [W]	Difference [W]
All	633	568	-65
L2 switch #1	160	157	QSFP: -3
Transponder #1	167	141	QSFP:-1 CFP2:-25
L2 switch #2	177	172	QSFP:-5
Transponder #2	129	98	QSFP:-2 CFP2:-29

4. Conclusion

We proposed an energy-saving method that uses traffic prediction to put unused DUs and optical transceivers to sleep and evaluated the power consumption reduction. In this evaluations, when one DU is put to sleep, the power consumption was reduced by 300 W for a DU and by 65 W for optical transceivers. The simulation results showed that a maximum of seven DUs can be put to sleep, achieving a total power reduction of 2555 W. From the above results, energy-saving of not only DUs but also transport is important, and the effectiveness of the proposed method was verified.

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